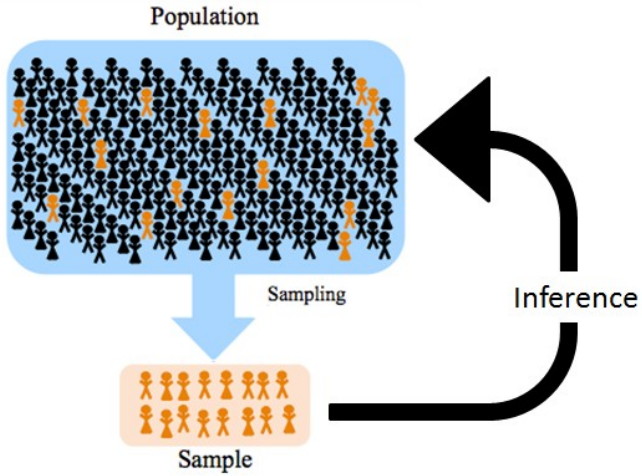
Lecture 22

Statistical Inference II



Announcements

- Course project feedback out later this week

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

Last Time

Statistical Inference

- Sampling
- Sample Variance

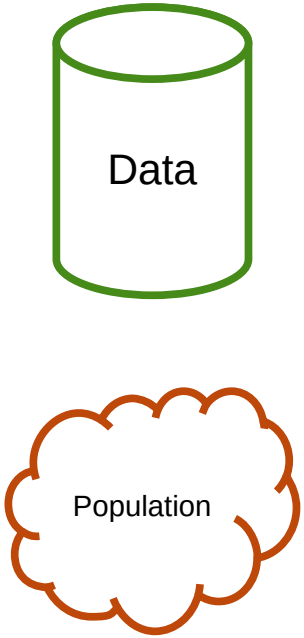
Today

Statistical Inference

- Statistical Hypotheses
- Difference of Means Test
- Testing other statistics

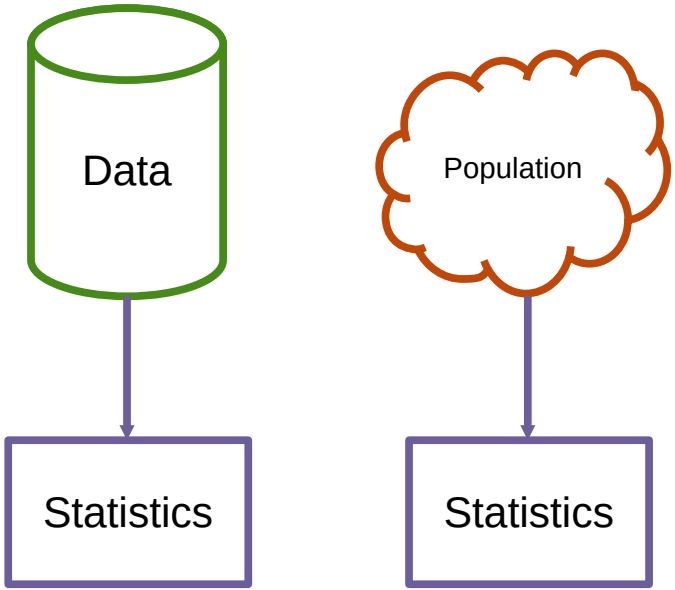
Samples & Populations

- For most of this course, we've worked "inside the box"
  - Focused on analyzing a dataset
  - Today, we're going to start reasoning about the population
- Think of our dataset as a **sample** from the population
  - A sample is often a concrete dataset
  - The population is often an idealized category
- Example
  - Sample: wait times at Meadville Starbucks collected today
  - Wait times during the morning at all Starbucks



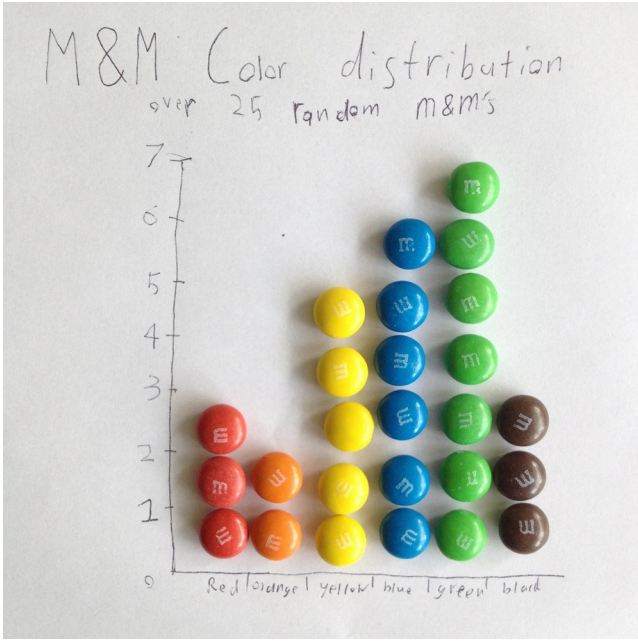
Statistics

- Summaries of our data
- We can learn a lot about a dataset through its statistics
- Populations have statistics too
- We can sometimes learn about the population statistics from our sample statistics
  - Called **inference**



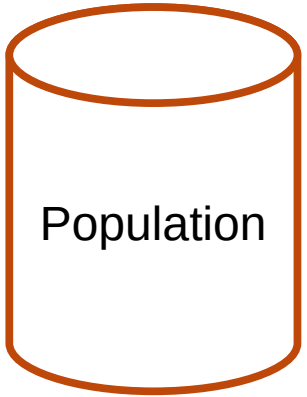
Sample Variation

- When you open a bag of M&Ms, do you get the exact same ratios every time?
- Even if the population ratio is uniform, under sampling, it's reasonable to expect some variation
- What do you predict would happen if we open lots of bags of M&Ms and looked at the color ratios?



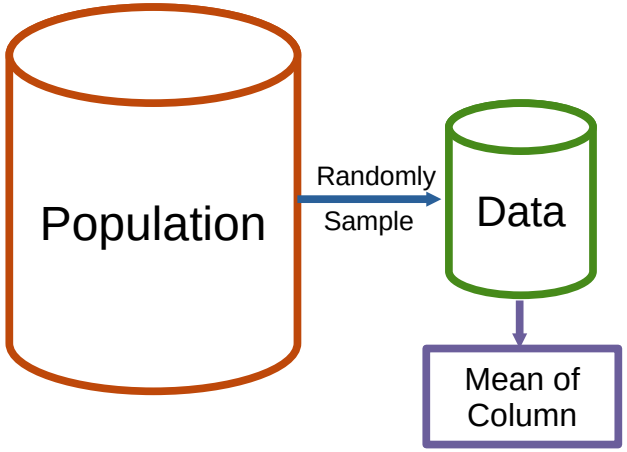
Sampling Variation

- Assume we have access to population
  - BIG assumption!



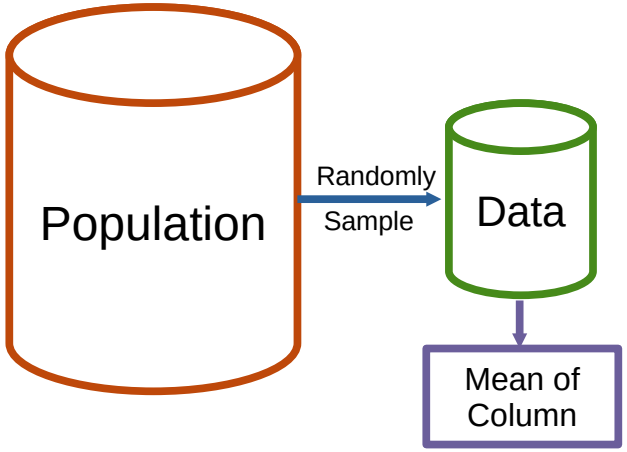
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- Obtain our dataset by randomly sampling some records from the population



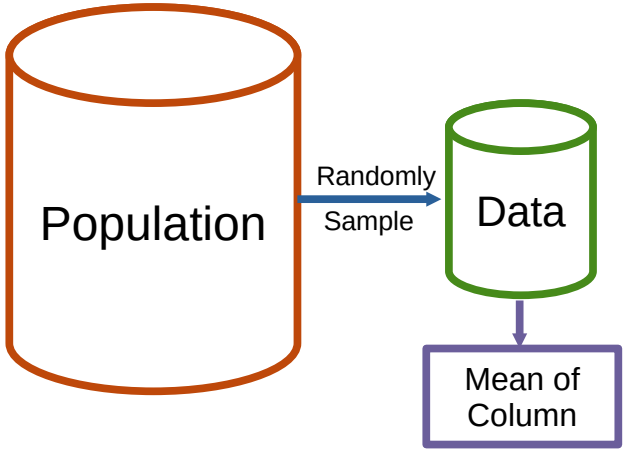
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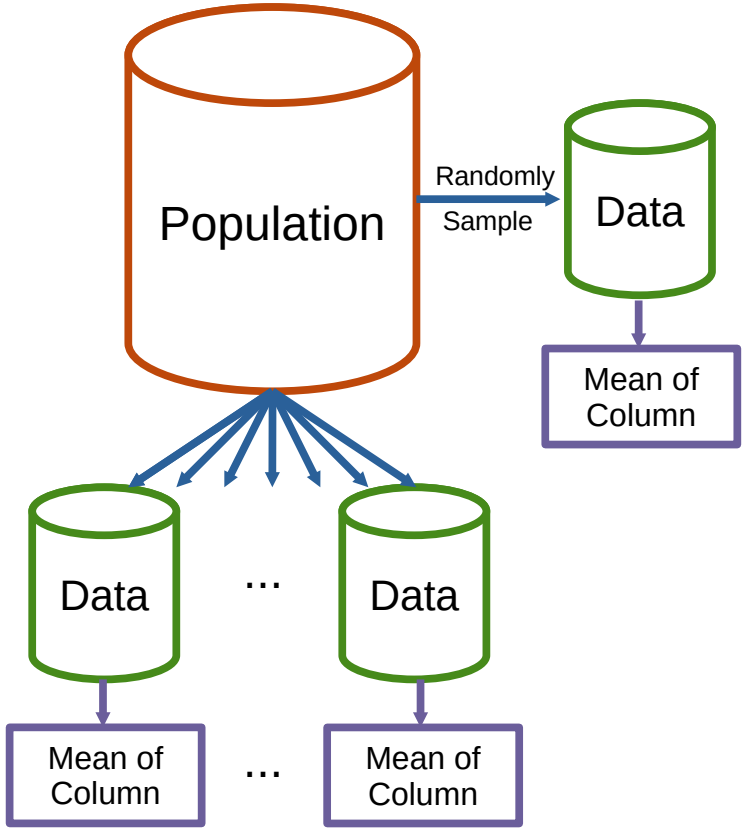
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  iii. Analyze distribution of statistics



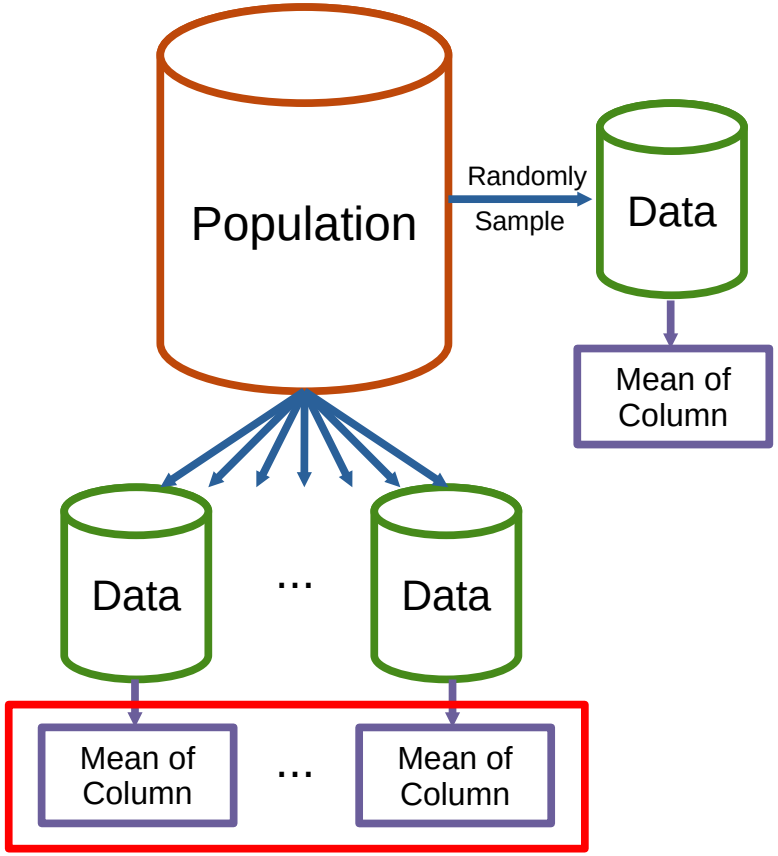
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Starbucks Dataset

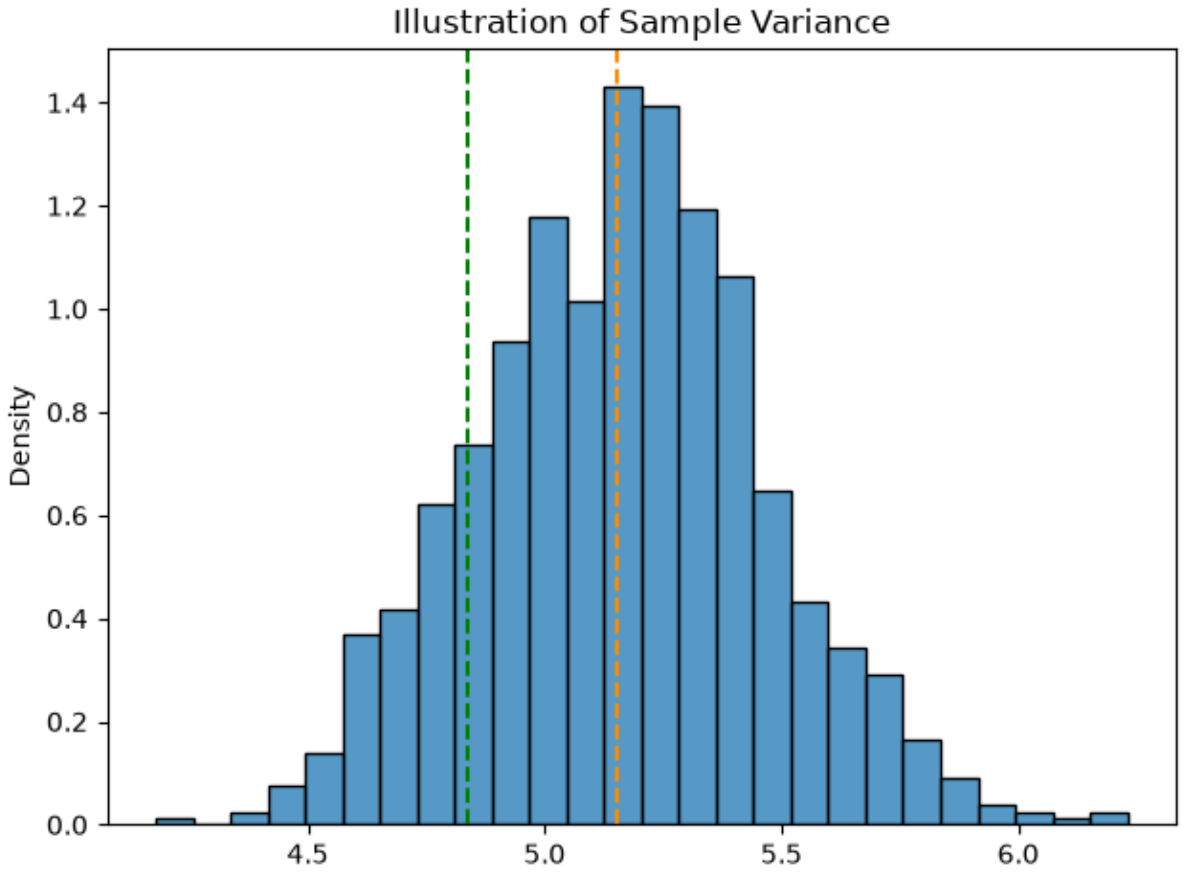
Average wait time

- Population: 5.15
- Sample: 4.82

Starbucks Dataset

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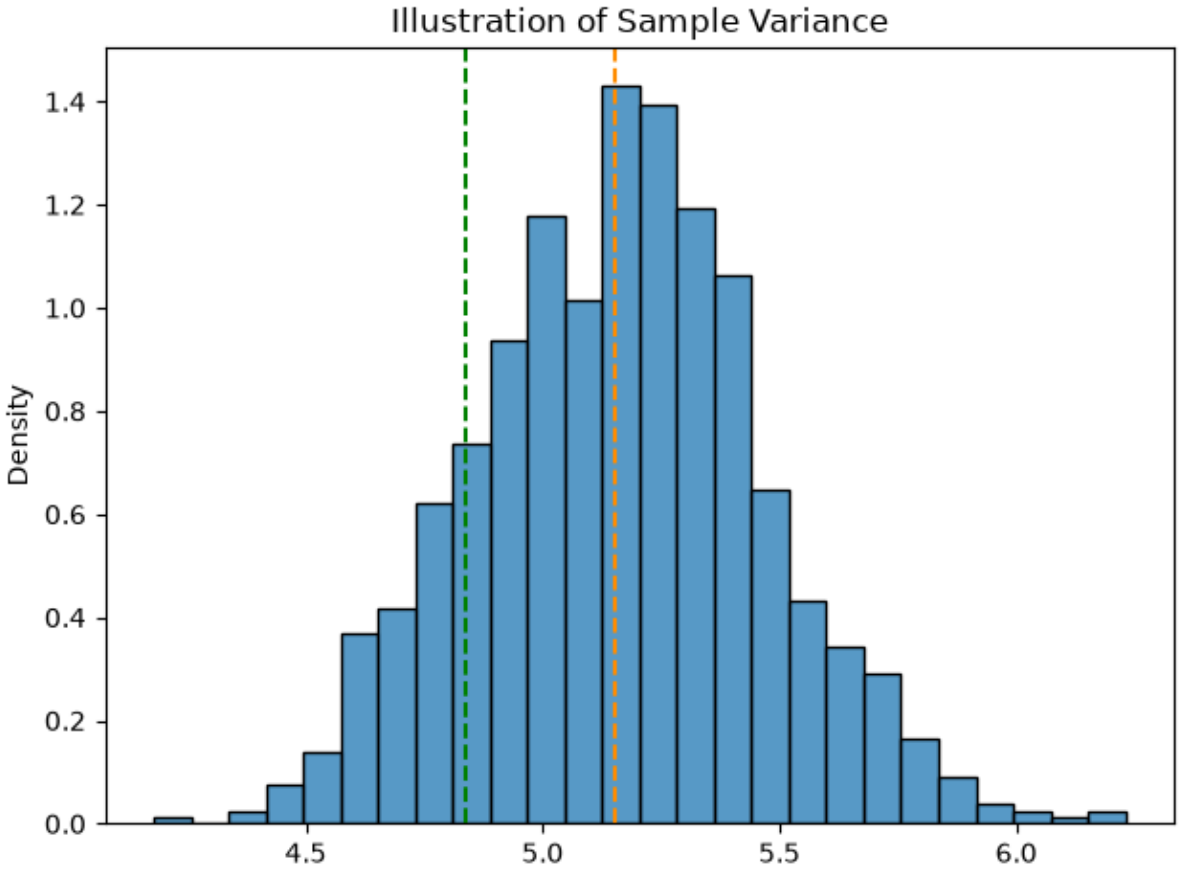
Starbucks Dataset

Average wait time

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Distribution of Sample Means

- Varies between about 4.5 and 6
- Mean of sampling distribution (5.16) approximates population mean (sampling is unbiased)



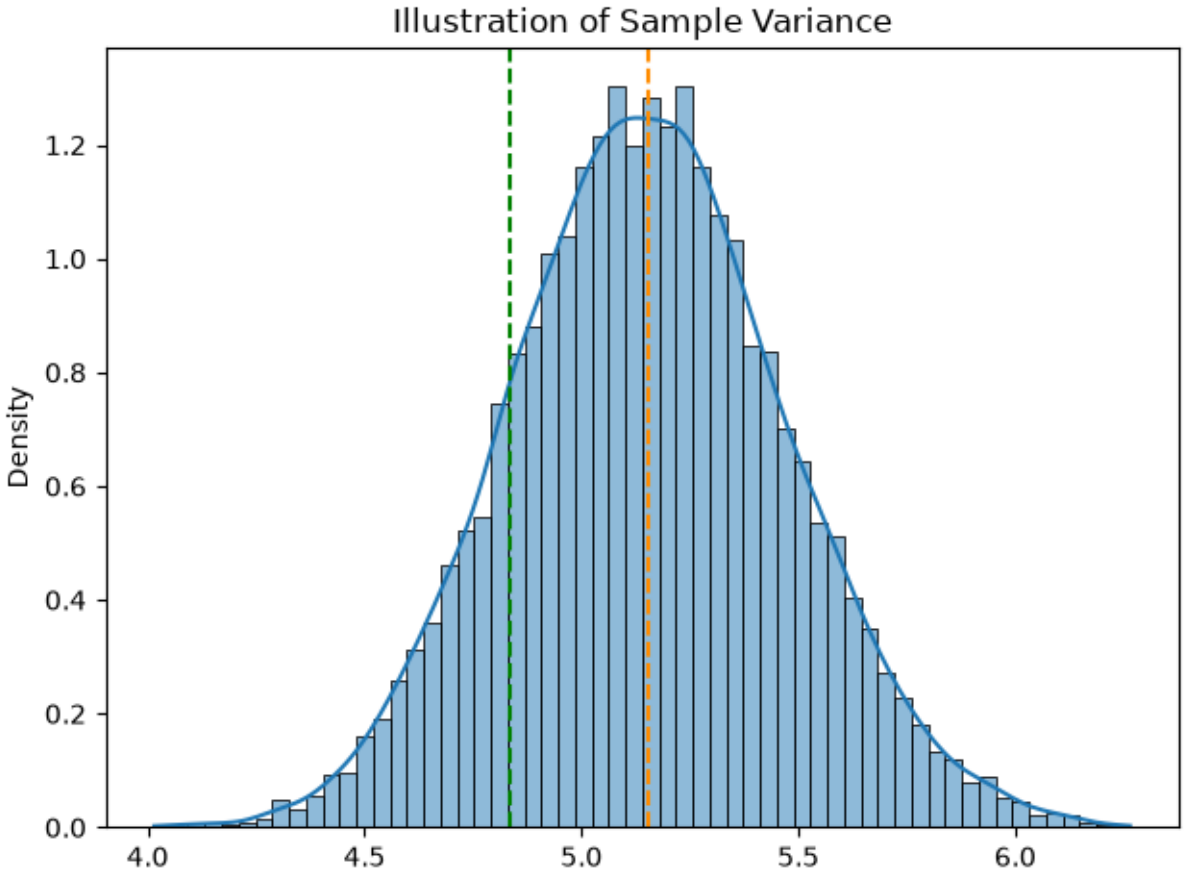
Starbucks Dataset

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Distribution of Sample Means

- Varies between about 4.5 and 6
- Mean of sampling distribution (5.16) approximates population mean (sampling is unbiased)
- Central limit theorem



Statistical Hypotheses

Ask: does a relationship in the sample hold for the population?

- Assume our sample was randomly sampled



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- Not necessarily, the observed relationship may be due to sampling variance



Statistical Hypotheses

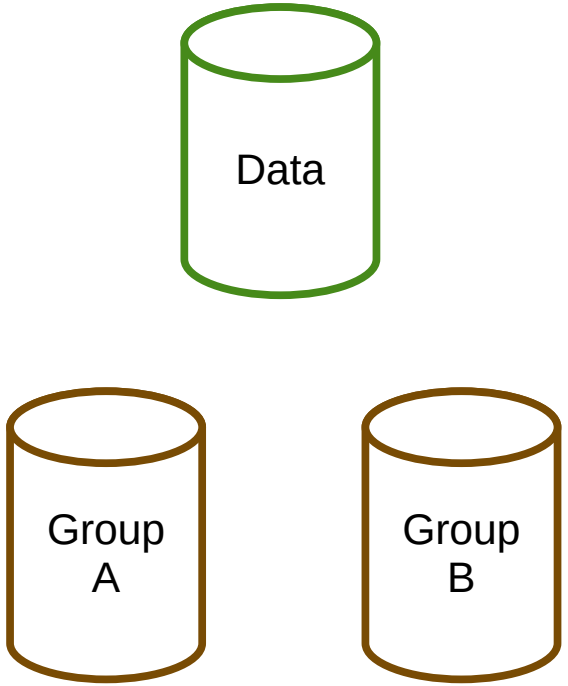
Ask: does a relationship in the sample hold for the population?

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Answer this by measuring whether or not the relationship differs from chance

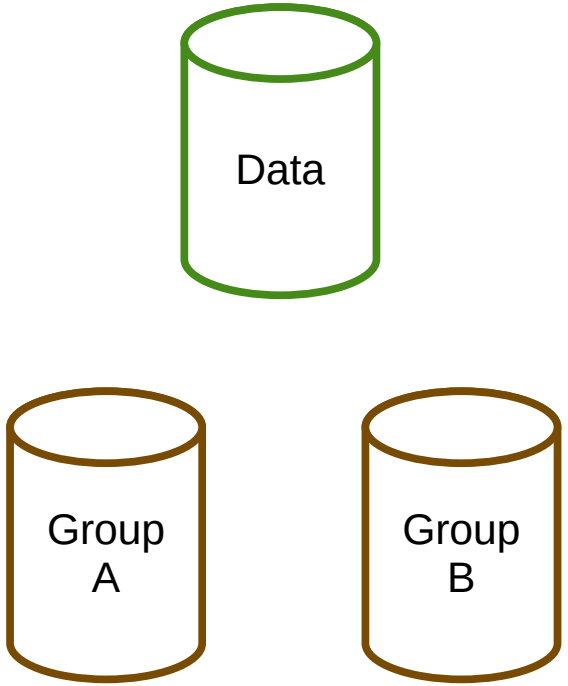
Subgroup Hypotheses



Subgroup Hypotheses

Null hypothesis: there's no difference between Group A and Group B

Alternative hypothesis: there is a difference between Group A and Group B

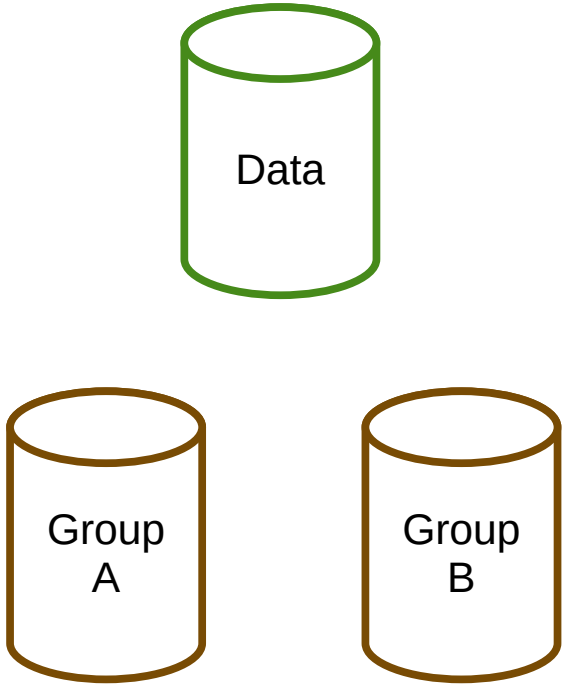


Subgroup Hypotheses

Null hypothesis: there's no difference between Group A and Group B

Alternative hypothesis: there is a difference between Group A and Group B

Goal: determine whether or not we should reject the null hypothesis



Mobile vs Walk-in Wait Times

Average wait times from our sample

- Mobile: 3.87
- Walk-in: 5.80

} Observed difference: 1.93

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Could this effect have arisen from chance?

- Is it an artifact produced by sampling variance?
- Is this effect real?

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Could this effect have arisen from chance?

- Is it an artifact produced by sampling variance?
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Can we figure this out without having access to the population?

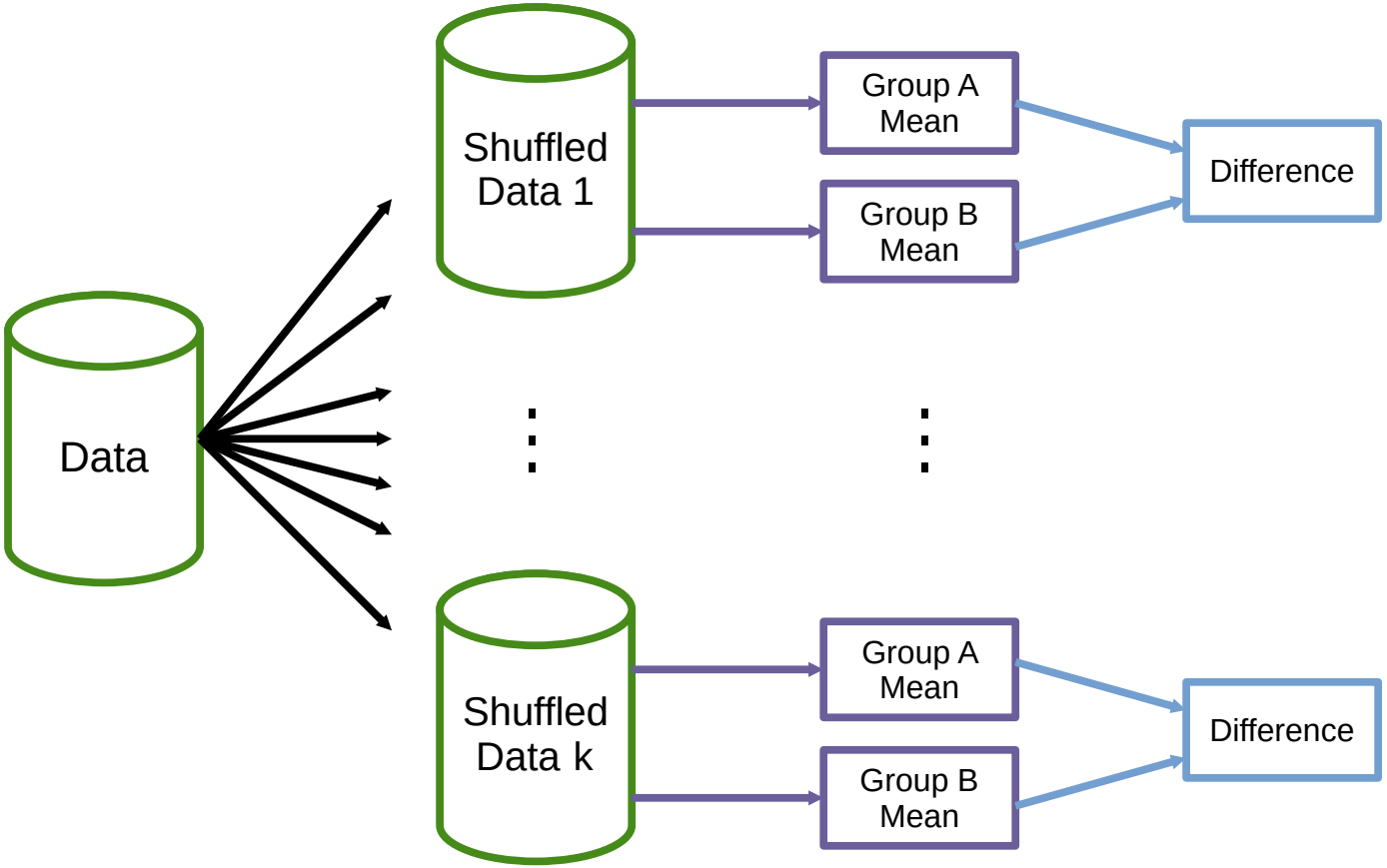
Permutation Test

Testing whether or not observed difference is due to chance

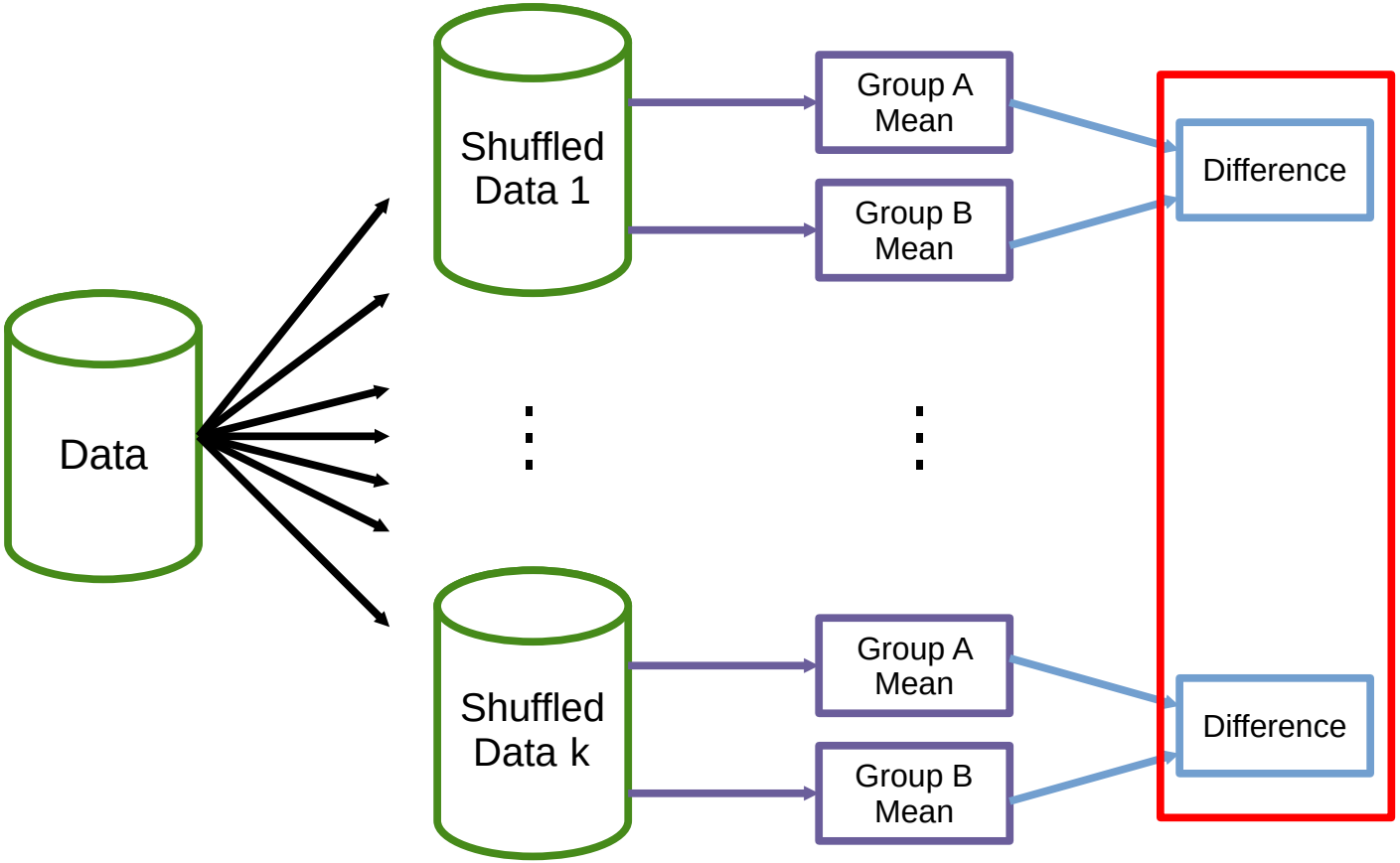
Idea:

i. Make k copies of our sample dataset (usually 1000)
ii. Randomly shuffle the order type labels on each new dataset
iii. Compute means of each order type for the new datasets
iv. Calculate percent of datasets where difference is greater than observed difference

Permutation Test



Permutation Test

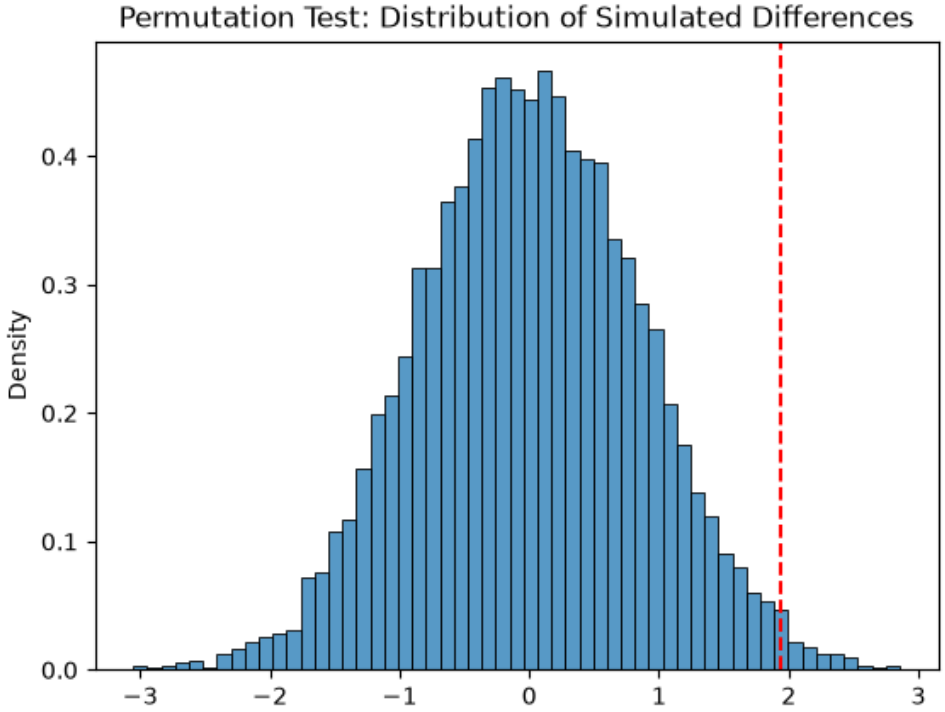


Analyze distribution of differences

Mobile vs Walk-in Wait Times

Average wait times from our sample

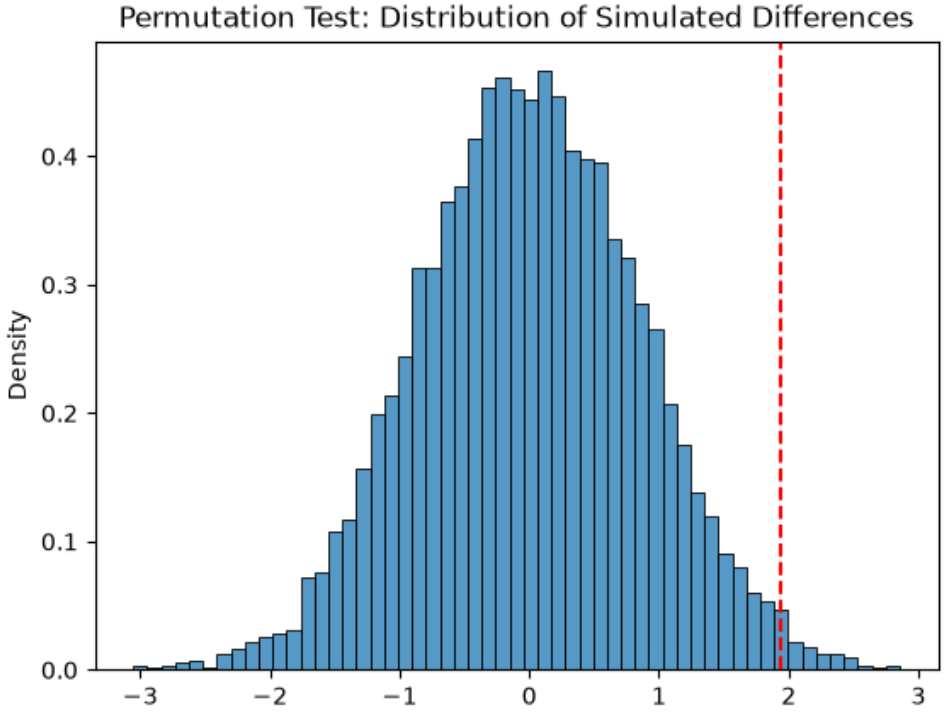
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- Walk-in: 5.80
- Difference: 1.93



Mobile vs Walk-in Wait Times

Only 1.14% of shuffled datasets produced a gap as big as the observed gap

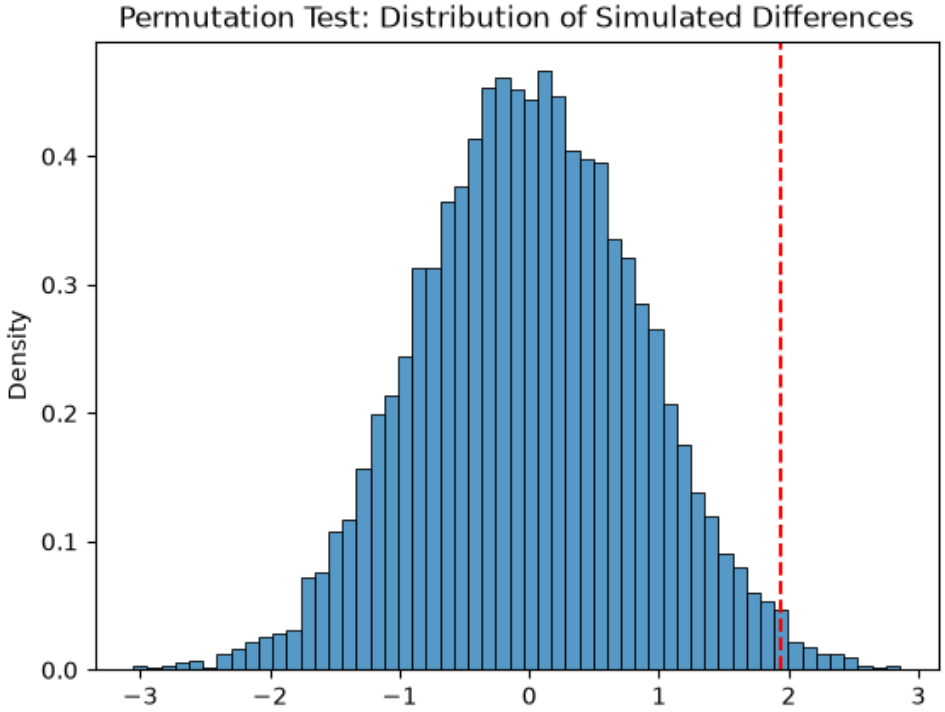
- If there were no differences between subgroups, the probability that an effect at least as large as the observed effect is approx. 0.0114



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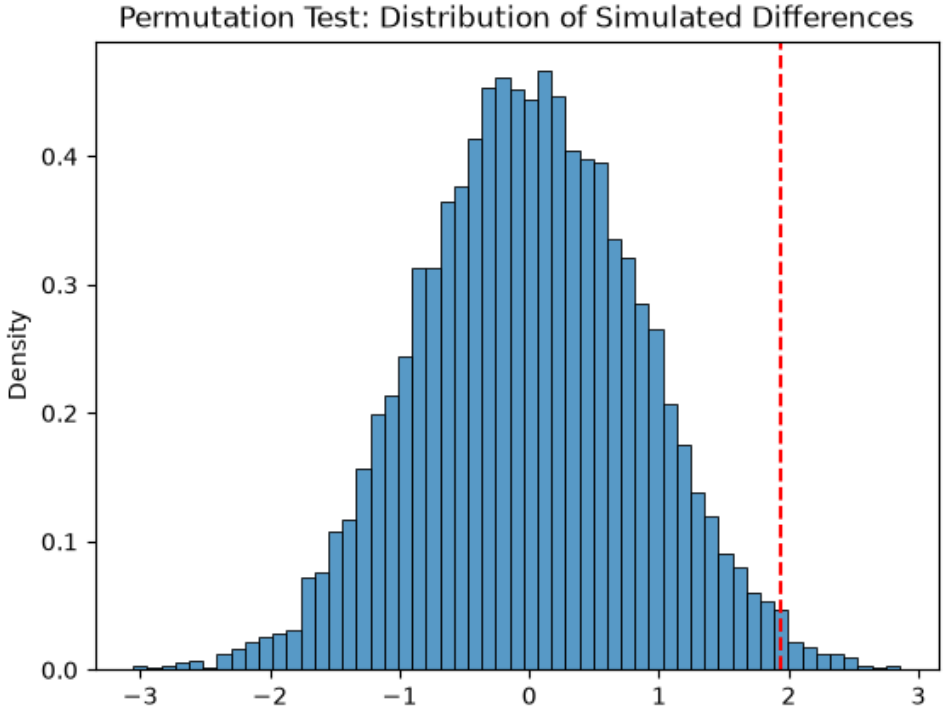
- If there were no differences between subgroups, the probability that an effect at least as large as the observed effect is approx. 0.0114
- We typically set the cut off for saying the effect is real at 0.05
- A version of a "p-value"



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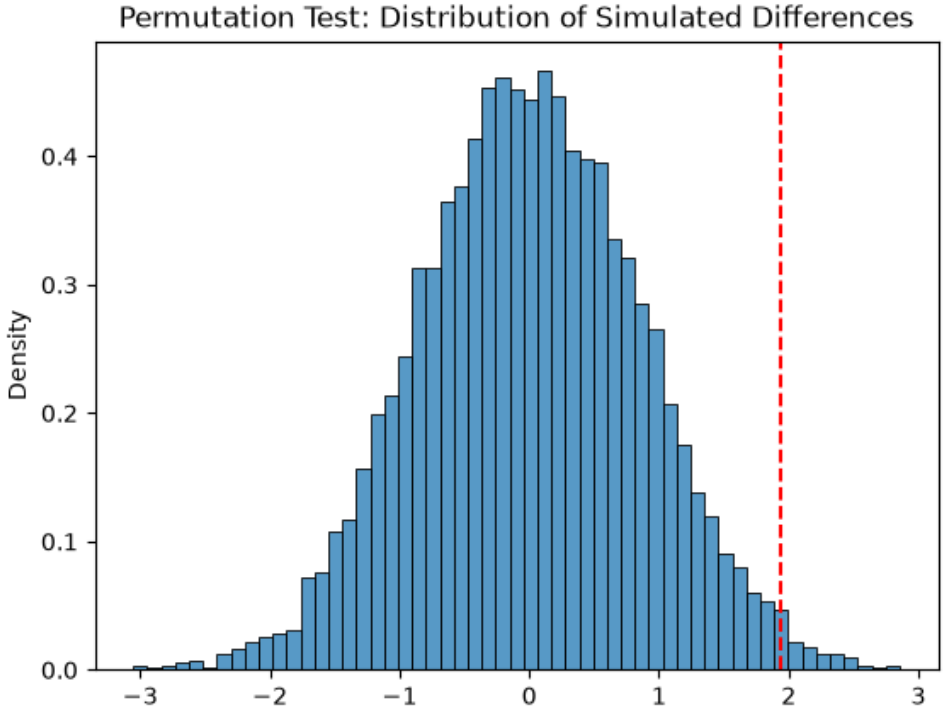
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- We typically set the cut off for saying the effect is real at 0.05
- A version of a "p-value"
- In this case, we would **reject** the null hypothesis
- Population effect: 1.90



Next Time

- Testing statistical hypotheses